THE PROJECTION FORMULA FOR DERIVED PUSHFORWARD ALONG A QUASI-COMPACT QUASI-SEPARATED MORPHISM

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Allowed standard prerequisites (editorial).

- The derived pushforward along a quasi-compact quasi-separated morphism commutes with direct sums on complexes with quasi-coherent cohomology sheaves. invoked at proof lines 5291-5293. Extends the assertion from single complexes to direct sums.
- On an affine base the tilde functor is an equivalence between D(A) and D_QCoh, and the P-resolution remark reduces assertions stable under direct sums and triangles to free modules. invoked at proof lines 5300-5306. Reduces the assertion to the shifts A[k] of the structure sheaf.

Source reference. Original equation 0B53: sources/stacks-chow/tags.txt, source lines 14021–14021. Cited at statement lines 5278-5280 as the map asserted to be an isomorphism; the proof of the assertion is inside the counted span.

Source reference. Original lemma 08DZ: sources/stacks-perfect/pinned-original-source.tex, source lines 803–803. Cited at proof lines 5292-5293 for the direct-sum stability; the stability argument is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 5301-5302 for the affine equivalence; the P-resolution reduction is inside the counted span.

Source reference. Original remark 09PB: sources/stacks-chow/tags.txt, source lines 12244–12244. Cited at proof lines 5305-5306 to reduce to the free case; the final conclusion is inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes. For E in $D_{QCoh}(\mathcal{O}_X)$ and K in $D_{QCoh}(\mathcal{O}_Y)$ the map*

$$Rf_*(E) \otimes^{\mathbf{L}}_{\mathcal{O}_Y} K \longrightarrow Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*K)$$

defined in Cohomology, Equation (0B53) is an isomorphism.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. To check the map is an isomorphism we may work locally on Y. Hence we reduce to the case that Y is affine.

Suppose that $K = \bigoplus K_i$ is a direct sum of some complexes $K_i \in D_{QCoh}(\mathcal{O}_Y)$. If the statement holds for each K_i, then it holds for K. Namely, the functors Lf^* and $\otimes^{\mathbf{L}}$ preserve direct sums by construction and Rf_* commutes with direct sums (for complexes with quasi-coherent cohomology sheaves) by Lemma 08DZ. Moreover, suppose that $K \to L \to M \to K[1]$ is a distinguished triangle in $D_{QCoh}(Y)$. Then if the statement of the lemma holds for two of K, L, M, then it holds for the third (as the functors involved are exact functors of triangulated categories).

Assume Y affine, say $Y = \mathrm{Spec}(A)$. The functor $^\sim : D(A) \to D_{QCoh}(\mathcal{O}_Y)$ is an equivalence (Lemma 06Z0). Let T be the property for $K \in D(A)$ that the statement of the lemma holds for $\widetilde{K}$. The discussion above and More on Algebra, Remark 09PB shows that it suffices to prove T holds for $A[k]$. This finishes the proof, as the statement of the lemma is clear for shifts of the structure sheaf. $\square$